DevSecOps Interview Q&A (AWS + Azure) – Detailed Version

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1. What is DevSecOps and why is it important?

Answer:
DevSecOps is an approach where security is integrated into every phase of the DevOps lifecycle, including development, build, testing, deployment, and operations.

Earlier, security was handled at the end of the development cycle, which caused delays and risks. DevSecOps solves this by making security a shared responsibility between developers, operations, and security teams.

In real-world projects, this helps in:

- Detecting vulnerabilities early
- Reducing production risks
- Automating security checks in CI/CD pipelines

In cloud environments like AWS and Azure, DevSecOps ensures that infrastructure, applications, and data are secure from the beginning.

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2. How do you implement DevSecOps in AWS?

Answer:
In AWS, DevSecOps is implemented by integrating security tools and practices into the CI/CD pipeline.

A typical flow looks like this:

- Developers push code to a repository
- Pipeline is triggered using AWS CodePipeline
- Build process runs in AWS CodeBuild

During the pipeline:

- Code is scanned for vulnerabilities (SAST tools)
- Dependencies are checked for known issues
- Docker images are built and scanned using Amazon ECR

After deployment:

- Infrastructure is scanned using Amazon Inspector
- Threat detection is handled by Amazon GuardDuty

Secrets like database passwords are securely stored in AWS Secrets Manager instead of hardcoding them.

This ensures security at every stage from code to production.

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3. How do you implement DevSecOps in Azure?

Answer:
In Azure, DevSecOps is implemented using integrated services for CI/CD, security monitoring, and compliance.

- Code pipelines are managed using Azure DevOps or GitHub Actions
- During the pipeline, security scans are added for code and dependencies

For infrastructure and runtime security:

- Microsoft Defender for Cloud continuously scans resources for vulnerabilities and misconfigurations

Sensitive information like secrets and certificates are stored securely in Azure Key Vault

Monitoring and logging are done using Azure Monitor to detect unusual activities.

This creates a secure and automated DevSecOps workflow in Azure.

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4. What is Shift Left in DevSecOps?

Answer:
Shift Left means moving security practices earlier in the software development lifecycle.

Instead of waiting until deployment or production, security checks are performed during:

- Coding
- Build stage
- Testing phase

For example:

- Running static code analysis while developers are writing code
- Scanning dependencies during the build

This reduces the cost and effort required to fix vulnerabilities because issues are caught early rather than in production.

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5. How do you secure a CI/CD pipeline?

Answer:
Securing a CI/CD pipeline involves protecting every stage from code commit to deployment.

First, at the source stage:

- Enforce branch protection
- Require code reviews
- Scan for secrets in repositories

During the build stage:

- Run SAST to detect code vulnerabilities
- Perform dependency scanning to identify vulnerable libraries

For containerized applications:

- Build Docker images
- Scan them using Amazon ECR or Azure Defender

Secrets must never be stored in code. Instead:

- Use AWS Secrets Manager or Azure Key Vault

Finally:

- Add security gates so the pipeline fails if vulnerabilities are detected

This ensures that only secure code is deployed.

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6. What is SAST and DAST? Explain with difference.

Answer:
SAST (Static Application Security Testing) analyzes source code without executing the application. It is used early in the development process to identify coding issues such as insecure functions or logic flaws.

DAST (Dynamic Application Security Testing) tests the application while it is running. It simulates real-world attacks to find vulnerabilities like SQL injection or cross-site scripting.

Key difference:

- SAST works on code and is used early
- DAST works on running applications and is used later

Both are important in a DevSecOps pipeline to ensure complete coverage.

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7. What is secrets management and why is it important?

Answer:
Secrets management is the process of securely storing and accessing sensitive information such as API keys, passwords, and tokens.

Hardcoding secrets in code or storing them in repositories is a major security risk.

In AWS:

- AWS Secrets Manager is used

In Azure:

- Azure Key Vault is used

These services allow:

- Secure storage
- Controlled access using IAM or RBAC
- Automatic rotation of secrets

This ensures that sensitive data is protected and managed securely.

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8. How do you secure Docker and container environments?

Answer:
Container security involves securing both the image and runtime environment.

For images:

- Use minimal base images like Alpine
- Remove unnecessary packages
- Scan images using Amazon ECR

For runtime:

- Avoid running containers as root
- Use read-only file systems where possible
- Apply resource limits

Also:

- Regularly update base images
- Use signed images to ensure integrity

This reduces the attack surface and improves container security.

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9. How do you secure Kubernetes (EKS or AKS)?

Answer:
Kubernetes security involves multiple layers.

For access control:

- Use RBAC
- Integrate IAM in EKS or Azure AD in AKS

For workloads:

- Run containers as non-root
- Use security contexts

For networking:

- Apply network policies to restrict communication

For images:

- Scan before deployment using Amazon ECR

For secrets:

- Use encrypted secrets with KMS or Key Vault

This ensures that both cluster and workloads are secure.

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10. What is vulnerability management lifecycle?

Answer:
Vulnerability management is a continuous process of identifying and fixing security issues.

Steps include:

1. Identification using scanners
2. Analysis to determine severity
3. Remediation by fixing or patching
4. Verification to ensure issue is resolved

In AWS:

- Amazon Inspector helps identify vulnerabilities

In Azure:

- Defender for Cloud performs similar tasks

This process ensures systems remain secure over time.

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11. What is IAM and how is it used in DevSecOps?

Answer:
Identity and Access Management controls who can access what resources.

In DevSecOps:

- Developers get limited access
- CI/CD pipelines use roles instead of credentials
- Services communicate using secure roles

Best practice:

- Always follow least privilege
- Avoid using root or admin access unnecessarily

This reduces the risk of unauthorized access.

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12. What is runtime security?

Answer:
Runtime security focuses on monitoring applications while they are running in production.

It helps detect:

- Unauthorized access
- Suspicious activity
- Malware or abnormal behavior

In AWS:

- Amazon GuardDuty is used

In Azure:

- Microsoft Defender for Cloud provides runtime protection

This ensures that threats are detected even after deployment.

13. What is “least privilege” and how do you implement it in AWS and Azure?

Answer:
Least privilege means giving users, applications, or services only the permissions they absolutely need to perform their tasks, and nothing more.

In real projects, over-permissioned access is one of the biggest security risks. If a service or user is compromised, excessive permissions can lead to major damage.

In AWS:

- You use IAM policies and roles
- For example, instead of giving full S3 access, you only allow read access to a specific bucket
- Services like EC2 or Lambda should use IAM roles instead of hardcoded credentials

In Azure:

- You use Role-Based Access Control (RBAC)
- Assign roles like Reader, Contributor, or custom roles at resource level

Best practice:

- Regularly audit permissions
- Remove unused access
- Use temporary credentials instead of long-term keys

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14. What is policy as code and how is it used in DevSecOps?

Answer:
Policy as code means defining security and compliance rules in code and enforcing them automatically.

Instead of manually checking configurations, policies are written and applied programmatically.

In AWS:

- You can use AWS Config to define rules like “S3 buckets should not be public”

In Azure:

- Azure Policy can enforce rules like “only allowed VM sizes” or “encryption must be enabled”

In DevSecOps pipelines:

- Policies are checked before deployment
- If a rule is violated, deployment is blocked

This ensures consistency, automation, and compliance across environments.

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15. What is “Defense in Depth” and how do you apply it in cloud?

Answer:
Defense in Depth is a security strategy where multiple layers of protection are applied so that if one layer fails, others still protect the system.

In cloud environments, this typically includes:

- Network layer
  Use security groups, NSGs, private subnets
- Application layer
  Use WAF to block attacks
- Identity layer
  Apply IAM roles and RBAC with least privilege
- Data layer
  Encrypt data at rest and in transit
- Monitoring layer
  Use tools like Amazon GuardDuty or Defender for Cloud

This layered approach significantly reduces risk in production systems.

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16. How do you secure Infrastructure as Code (Terraform/CloudFormation)?

Answer:
Infrastructure as Code can introduce risks if not secured properly, such as exposing resources publicly or misconfigurations.

To secure IaC:

- Scan code before deployment using tools like Checkov or tfsec
- Avoid hardcoding secrets in Terraform files
- Store secrets in:
    - AWS Secrets Manager
    - Azure Key Vault
- Use remote backend for state files:
    - AWS S3 with encryption and DynamoDB locking
- Restrict access to state files because they may contain sensitive data
- Enable version control and review changes before applying

This ensures infrastructure is secure before it is even created.

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17. What is container image scanning and why is it important?

Answer:
Container image scanning checks Docker images for vulnerabilities such as outdated libraries or known security issues.

If you deploy unscanned images, you might introduce critical vulnerabilities into production.

In AWS:

- Amazon ECR provides built-in image scanning

In Azure:

- Defender for Cloud scans container images

Best practices:

- Scan images during build stage
- Fail pipeline if critical vulnerabilities are found
- Regularly update base images

This ensures only secure images are deployed.

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18. How do you manage secrets securely in a CI/CD pipeline?

Answer:
Secrets should never be stored in source code, configuration files, or Docker images.

Instead:

In AWS:

- Use AWS Secrets Manager

In Azure:

- Use Azure Key Vault

In pipelines:

- Inject secrets at runtime using environment variables
- Use short-lived tokens instead of static credentials
- Rotate secrets regularly

Also:

- Restrict access using IAM or RBAC
- Audit usage of secrets

This ensures sensitive data is not exposed.

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19. What is runtime security and how is it different from build-time security?

Answer:
Build-time security focuses on identifying vulnerabilities during the CI/CD process before deployment.

Examples:

- SAST
- Dependency scanning
- Image scanning

Runtime security focuses on detecting threats when the application is running in production.

Examples:

- Unusual network activity
- Unauthorized access attempts

In AWS:

- Amazon GuardDuty monitors runtime threats

In Azure:

- Defender for Cloud provides runtime protection

Both are important because some threats only appear after deployment.

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20. How do you secure APIs in AWS and Azure?

Answer:
APIs are a common attack surface, so they must be secured properly.

Key steps:

- Authentication
  Use OAuth or JWT tokens
- Authorization
  Use IAM roles or RBAC
- Encryption
  Use HTTPS/TLS
- Rate limiting
  Prevent abuse or DDoS attacks

In AWS:

- API Gateway + WAF

In Azure:

- API Management + WAF

Also:

- Validate input data
- Log API requests for monitoring

This ensures APIs are protected from common attacks.

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21. What is logging and monitoring in DevSecOps?

Answer:
Logging and monitoring help track system behavior and detect security incidents.

In AWS:

- CloudTrail logs all API activity
- CloudWatch monitors application logs and metrics

In Azure:

- Azure Monitor and Log Analytics provide similar functionality

Use cases:

- Detect unauthorized access
- Track changes in infrastructure
- Investigate incidents

Without logging, it is impossible to perform proper security analysis.

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22. What is incident response in DevSecOps?

Answer:
Incident response is the process of handling security incidents in a structured way.

Steps include:

1. Detection
   Identify the issue using monitoring tools
2. Containment
   Isolate affected systems
3. Eradication
   Remove the threat
4. Recovery
   Restore systems to normal state

5. Post-analysis
   Identify root cause and improve security

In cloud:

- Use alerts from GuardDuty or Defender
- Automate response using Lambda or Logic Apps

This reduces impact and prevents future incidents.

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23. What is compliance and how do you enforce it?

Answer:
Compliance ensures systems follow standards like ISO, SOC2, or GDPR.

In AWS:

- AWS Config checks resource configurations

In Azure:

- Azure Policy enforces compliance rules

Examples:

- Ensure encryption is enabled
- Prevent public access to storage

In DevSecOps:

- Compliance checks are automated in pipelines
- Non-compliant resources are blocked

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24. What is a real-world DevSecOps pipeline example?

Answer:
A real pipeline includes multiple security layers.

Flow:

- Developer pushes code
- Pipeline starts using AWS CodePipeline
- Build runs in AWS CodeBuild

Security steps:

- SAST scan
- Dependency scan
- Build Docker image

- Scan image using Amazon ECR

Deployment:

- Deploy to Kubernetes (EKS or AKS)

Post-deployment:

- Monitor using Amazon GuardDuty

If any vulnerability is found, pipeline fails and deployment is blocked.

25. What is Cloud Security Posture Management (CSPM) and how is it used?

Answer:
Cloud Security Posture Management is used to continuously monitor cloud resources for misconfigurations and compliance violations.

In real environments, many security issues are not due to attacks but due to incorrect configurations such as:

- Public storage buckets
- खुले ports
- Missing encryption

In AWS:

- AWS Security Hub aggregates findings from multiple services like GuardDuty and Inspector

In Azure:

- Microsoft Defender for Cloud provides similar capabilities

CSPM tools:

- Continuously scan resources
- Provide alerts for risks
- Suggest remediation steps

This helps maintain a secure cloud environment without manual checks.

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26. What is the role of DevSecOps in a microservices architecture?

Answer:
In microservices, applications are broken into multiple smaller services, each with its own deployment lifecycle.

This increases complexity and also increases the attack surface.

DevSecOps helps by:

- Securing each microservice independently
- Enforcing authentication between services (service-to-service security)
- Using API gateways for centralized security
- Scanning each container image before deployment

In Kubernetes environments:

- Apply network policies to restrict communication
- Use RBAC for access control
- Monitor each service at runtime

Without DevSecOps, managing security in microservices becomes very difficult.

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27. How do you secure communication between microservices?

Answer:
Service-to-service communication must be secured to prevent unauthorized access.

Common approaches:

- Use TLS encryption for all communication
- Implement mutual TLS (mTLS) so both client and server verify each other
- Use service mesh tools like Istio

In cloud:

- Use private networking (VPC or VNet)
- Restrict communication using security groups or NSGs

Authentication:

- Use tokens like JWT

This ensures that only authorized services can communicate.

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28. What is a security gate in CI/CD and how do you implement it?

Answer:
A security gate is a checkpoint in the pipeline that prevents insecure code from moving forward.

For example:

- If SAST finds critical vulnerabilities, the build fails

- If a Docker image has high-risk CVEs, deployment is blocked

Implementation:

- Add scanning tools in pipeline stages
- Define thresholds (for example, no high severity vulnerabilities allowed)

In AWS:

- Integrate scans in AWS CodePipeline

In Azure:

- Use policies in Azure DevOps pipelines

This ensures that insecure code never reaches production.

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29. What is threat modeling and why is it important?

Answer:
Threat modeling is the process of identifying potential security threats during the design phase of an application.

Instead of reacting to attacks, you proactively think about:

- What can go wrong
- How attackers might exploit the system
- How to mitigate those risks

Example:

- Identifying that an API might be exposed publicly and needs authentication

Common model:

- STRIDE (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege)

This helps design secure systems from the beginning.

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30. How do you handle a security incident in production?

Answer:
Handling a security incident requires a structured approach.

Steps:

1. Detection
   Alerts from monitoring tools like Amazon GuardDuty
2. Containment
   Isolate affected resources (for example, stop compromised EC2 instance)

3. Investigation
   Analyze logs using CloudTrail or Azure Monitor
4. Remediation
   Fix vulnerability, patch system
5. Recovery
   Restore services
6. Post-incident review
   Identify root cause and improve processes

In DevSecOps, many of these steps can be automated.

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31. What is SBOM and why is it important in DevSecOps?

Answer:
SBOM (Software Bill of Materials) is a detailed list of all components, libraries, and dependencies used in an application.

It is important because:

- Helps track vulnerable components
- Improves transparency
- Required for compliance in many organizations

Example:
If a vulnerability is discovered in a library, SBOM helps quickly identify if your application is affected.

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32. What is the difference between IAM roles and policies?

Answer:
In AWS:

- Policy
  Defines permissions (what actions are allowed or denied)
- Role
  An identity that can be assumed by users or services

Example:

- A role may have a policy that allows access to S3

In Azure:

- Similar concept using RBAC roles and role assignments

Roles are used instead of hardcoded credentials for better security.

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33. What is drift detection and why is it important?

Answer:
Drift detection identifies changes between the desired configuration and the actual state of infrastructure.

Example:

- Terraform defines a private S3 bucket
- Someone manually makes it public

Drift detection tools:

- AWS Config
- Terraform plan

This ensures that unauthorized changes are detected and corrected.

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34. How do you secure serverless applications?

Answer:
Serverless applications also need strong security controls.

Key practices:

- Use IAM roles with least privilege
- Validate input to prevent attacks
- Store secrets in:
    - AWS Secrets Manager
    - Azure Key Vault
- Enable logging and monitoring
- Avoid long-running permissions

This ensures that serverless functions remain secure.

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35. What is a common DevSecOps failure in real projects?

Answer:
One common failure is integrating DevOps but ignoring security.

Example scenario:

- Fast CI/CD pipeline

- No vulnerability scanning
- Secrets stored in code

Result:

- Application deployed with vulnerabilities
- High risk of breach

Fix:

- Add security checks at every stage
- Enforce security gates
- Train developers on secure coding

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36. How do you secure data in cloud environments?

Answer:
Data security involves protecting data at all stages.

At rest:

- Use encryption (S3, EBS, Azure Storage)

In transit:

- Use HTTPS/TLS

Access control:

- IAM roles and RBAC

Monitoring:

- Track access logs

Backup:

- Enable versioning and backups

This ensures data is protected from unauthorized access and loss.

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37. What is Zero Trust architecture in cloud?

Answer:
Zero Trust means no user or system is trusted by default, even if inside the network.

Principles:

- Verify identity continuously
- Use least privilege

- Monitor all activity

In practice:

- Use strong authentication
- Restrict network access
- Monitor behavior

This reduces risk from internal and external threats.

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38. How do you integrate security in Kubernetes CI/CD pipelines?

Answer:
Security must be integrated before deployment.

Steps:

- Scan code and dependencies
- Scan container images using Amazon ECR
- Validate Kubernetes manifests
- Enforce policies (no privileged containers)

Deployment:

- Apply RBAC
- Use network policies

Post-deployment:

- Monitor cluster activity

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39. What is the role of WAF in DevSecOps?

Answer:
Web Application Firewall protects applications from common attacks.

It filters incoming traffic and blocks:

- SQL injection
- Cross-site scripting

In AWS:

- AWS WAF

In Azure:

- Azure WAF

It acts as a security layer before requests reach the application.

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40. How do you design a secure multi-tier architecture in AWS or Azure?

Answer:
A secure architecture separates components into layers.

Typical design:

- Presentation layer
  Load balancer in public subnet
- Application layer
  Application servers in private subnet
- Database layer
  Database in private subnet with no internet access

Security controls:

- Use security groups or NSGs
- Enable encryption
- Restrict access using IAM/RBAC
- Monitor using security tools

This ensures high availability and strong security.